EXP: 12
DATE: 8/10/26

End-End Communication at Transport Layer.

AIM:

To implement TCP and UDP echo client-server communication and a chat program using socket programming.

TCP Echo Client-Server - tcp-echo.py

```
import socket
import threading
def start_server():
    server = socket.socket(socket.AF_INET,
socket.SOCK_STREAM)
    server.bind(("127.0.0.1", 5000))
    server.listen(1)
    print("[SERVER] Listening on
127.0.0.1:5000")
    conn, addr = server.accept()
    print("[SERVER] connected by", addr)
    while True:
        data = conn.recv(1024).decode()
        if not data or data.lower() ==
"quit":
            print("[SERVER] Connection
closed by client.")
            break
        print("[SERVER] Received:",data)
        conn.send(("Server received :"+
data).encode())
```

```
        conn.close()
    server.close()

def start_client():
    client = socket.socket(socket.AF_INET,
socket.SOCK_STREAM)
    client.connect(("127.0.0.1", 5000))
    print("[CLIENT] Connected to server")
    while True:
        msg = input("Enter message (or
'quit' to exit): ")
        client.send(msg.encode())
        if msg.lower() == "quit":
            break
        response = client.recv(1024).decode()
        print("[SERVER RESPONSE]", response)
    client.close()

if __name__ == "__main__":
    server_thread = threading.Thread(
        target = start_server, daemon=
        TRUE)
        server
    server_thread.start()
    start_client()
```

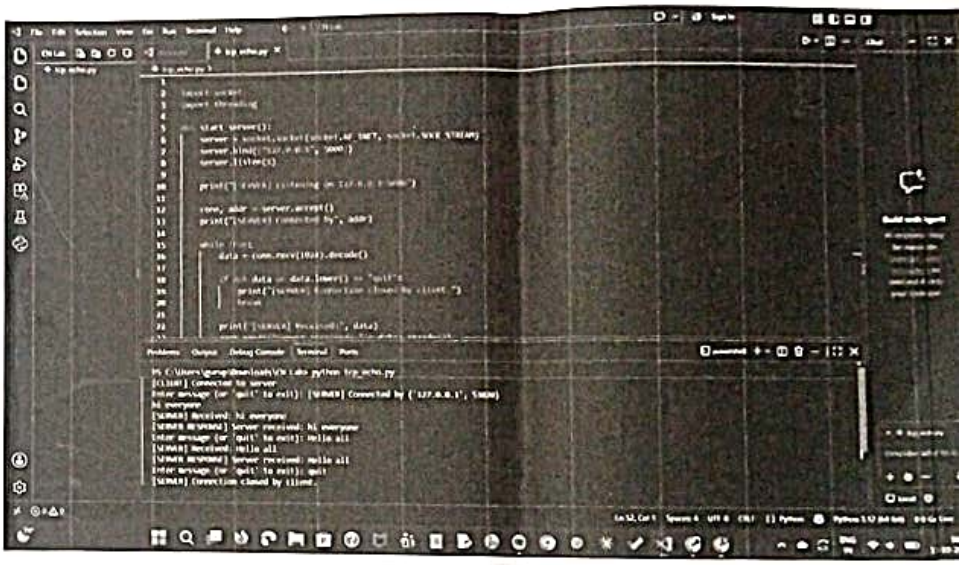
UDP Echo client – Server:

udp_echo.py

```
import socket
import threading
def udp_server():
    server = socket.socket(socket.AF_INET
socket.SOCK_DGRAM)
    server.bind(("127.0.0.1", 12000))
    print("[SERVER] Running on
127.0.0.1:12000")
    while True:
        data, addr = server.recvfrom(1024)
        msg = data.decode()
        if msg.lower() == "quit":
            print("[SERVER] client",
addr, " ended communication.")
            break
        Print("[SERVER] Received from",
addr, ":", msg)
        server.sendto(data, addr)
    server.close()
    print("[SERVER] Stopped.")
def udp_client():
    client = socket.socket(socket.AF_INET
, socket.SOCK_DGRAM)
    Server_address = ("127.0.0.1", 12000)
```

```
        while True:
            msg = input("Enter message (or 'quit' to stop): ")
            client.sendto(msg.encode(), server_address)
            if msg.lower() == "quit":
                print("[CLIENT] Stopped.")
                break
            data, _ = client.recvfrom(1024)
            print("Echo from Server :", data.decode())
        client.close()

if __name__ == "__main__":
    threading.Thread(target=udp_server, daemon=True).start()
    udp_client()
```



RESULT:

The TCP and UDP echo client-server programs were implemented and executed successfully.

